

Branching Logic

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the tested path for Branching and boundary tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Branching and boundary tests. Proposed explanation: “Every condition in the chain always runs”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 branch is placed first?

6. Solve the given case

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the boundary.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 3 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Branch when a reader has a hard constraint such as language, format, or maximum length.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 branch is placed first?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the chain

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented branches.

My reasoning and evidence

13. W2 Trace the tested path

For -1, 12 and 13, record checked conditions, skipped conditions and the final label. Run each as a fresh script after predicting.

My reasoning and evidence

14. W3 Repair the overwritten result

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

My reasoning and evidence

15. W4 Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an ordered if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My reasoning and evidence

16. W5 Choose revealing regression tests

A repair changes the small-count condition from < 6 to ≤ 6 . Name the case that exposes the original error and two different kinds of earlier cases to rerun.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Branching Logic

A fictional reading-time labeler uses one Python if/elif/else chain.

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as `minutes < 0`, `minutes == 0`, `minutes <= 10`, otherwise. Print the assigned label once.

AI ACADEMY / AI-2/3

Branching Logic

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Negative check

-1 takes Invalid and skips later branches.



2. Zero check

0 takes Empty, not Short.



3. Inclusive short range

1 and 10 take Short.



4. Remaining inputs

11 takes Long; only the selected branch assigns the label.

CHANGE ONE THING

Move `minutes <= 10` to the first branch.

Explain the changed result using the matching lesson.

Give expected labels for -1, 0, 10, 11 and identify the equality boundary for Short.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-time labeler uses one Python if/elif/else chain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 11 takes Long; only the selected branch assigns the label.

1. Negative check
2. Zero check
3. Inclusive short range
4. Remaining inputs

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Remaining inputs 11 takes Long; only the selected branch assigns the label. The third comparison alone would include negative values and zero, but earlier branches already handle them. An elif chain selects its first true branch. Multiple independent if statements would have different behavior and could overwrite an earlier label. Boundary traces show why

order belongs to the specification.

Practice 2: explain the deciding evidence

Changed-case practice: Move `minutes <= 10` to the first branch. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. `-1` and `0` wrongly receive `Short`.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: `-1` and `0` wrongly receive `Short`. Both satisfy the earlier comparison before their intended checks are reached.

Repair a missing building block

Review `Decompose a Goal` (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores `needed` but omits `print(needed)?`. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use `Download completed work` in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
